

Problem Set 6

1. Conv1d and the sliding dot product

Given the DSP convolution sum $y_{DSP}[n] = \sum_k h[k]x[n - k]$ and PyTorch's cross-correlation convention $y_{torch}[n] = \sum_k w[k]x[n + k]$:

- (a) Derive the relationship $w[k] = h[M - k]$ (where h has $M + 1$ taps, indices $0, \dots, M$) that makes `F.conv1d` reproduce the textbook causal FIR convolution, and state the required left-padding amount.
- (b) Implement both, the NumPy convolution `np.convolve` and the equivalent `F.conv1d` computation for a Hamming-windowed FIR lowpass filter with $\omega_c = \pi/3$ and length $N = 51$ on a chirp signal (`chirp_001.wav`). Report the maximum absolute difference between the two outputs (you should get something on the order of 10^{-6} to 10^{-7} in float32). Show evidence that your filter works as desired (e.g., by recording the filtered chirp signal as .wav-file and using a program like Audacity to generate a spectrogram). Show all your Python and PyTorch code in your solution.
- (c) A colleague says “PyTorch's `conv1d` does convolution backwards, so it's a bug.” Give a one-paragraph rebuttal, using the ML viewpoint discussed in lecture.

2. Pole-zero geometry

For

$$H(z) = \frac{1 - 0.8z^{-1}}{1 + 0.5z^{-1}}$$

- (a) Identify the zero and the pole.
- (b) Explain geometrically why $|H(e^{j\omega})|$ is the product of zero-vector lengths divided by the product of pole-vector lengths.
- (c) Predict qualitatively whether the magnitude is larger near $\omega = 0$ or $\omega = \pi$.

3. Phase delay and group delay

Let $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\phi(\omega)}$ with $\phi(\omega) = -3\omega + 0.4\sin(\omega)$.

(a) Compute the phase delay

$$\tau_p(\omega) = -\frac{\phi(\omega)}{\omega}$$

for $\omega \neq 0$.

(b) Compute the group delay

$$\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega}.$$

(c) Explain why the system is not linear phase.

4. First order allpass

For real $|a| < 1$,

$$H_{ap}(z) = \frac{z^{-1} - a}{1 - az^{-1}}.$$

Show that

$$|H_{ap}(e^{j\omega})| = 1$$

for all ω . Locate the pole and zero and explain the reciprocal relationship.

5. Inverse system properties (ECEN 5632 Problem)

If the system function $H(z)$ of an LTI system has a pole-zero plot as shown on the right and the system is causal, can the inverse system $H_i(z)$, where $H(z)H_i(z) = 1$, be both causal and stable? Clearly justify your answer.

